

Practical 1.

Scores obtained by students appeared in an entrance examination of B Sc CSIT from a college are as follows;

78, 56, 45, 39, 12, 72, 45, 56, 44, 32, 61, 58, 34, 21, 10, 61, 53, 68, 77, 13, 26, 31, 42, 55

Determine

- Mean, median, mode, Geometric mean, Harmonic mean, minimum, maximum, quartiles
- Range, Coefficient of range, interquartile range, semi interquartile range (quartile deviation), coefficient of quartile deviation, standard deviation, coefficient of standard deviation, variance, coefficient of variation
- Coefficient of skewness using quartiles, coefficient of skewness using moments, coefficient of skewness using box and whisker plot, coefficient of kurtosis using percentiles, coefficient of kurtosis using moments

Working formula:

Let x be a variable taking different values

$$\text{Mean}(\bar{x}) = \frac{\sum x}{n}$$

Median = $(n + 1)/2^{th}$ item of data after arranging data in ascending order or descending order

Mode = most repeated value of x

$$\text{Geometric mean} = \text{Antilog} \left(\frac{\sum \log x}{n} \right)$$

$$\text{Harmonic mean} = n / \sum \frac{1}{x}$$

Maximum = maximum value of x

Minimum = minimum value of x

Quartiles (Q_i) = $i(n + 1)/4^{th}$ item of data after arranging data in ascending order, $i = 1, 2, 3$

$$\text{Range} = X_{\max} - X_{\min}$$

$$\text{Coefficient of range} = (X_{\max} - X_{\min}) / (X_{\max} + X_{\min})$$

$$\text{Interquartile range} = (Q_3 - Q_1)$$

$$\text{Semi interquartile range} = (Q_3 - Q_1)/2$$

$$\text{Coefficient of quartile deviation} = (Q_3 - Q_1)/(Q_3 + Q_1)$$

$$\text{Standard deviation } (\sigma) = \sqrt{\frac{1}{n} \sum (x - \bar{x})^2} = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$$\text{Coefficient of SD} = \frac{\sigma}{\bar{x}}$$

$$\text{Variance } (\sigma^2) = SD^2$$

$$\text{Coefficient of variation (CV)} = \frac{\sigma}{\bar{x}} * 100\%$$

$$\text{Coefficient of skewness using quartiles } (S_{KB}) = \frac{Q_3 + Q_1 - 2Md}{Q_3 - Q_1}$$

$$\text{Coefficient of skewness using moments } (Y_1) = \frac{\mu_3}{\mu_2^{3/2}}$$

$$\mu_3 = \frac{\sum (x - \bar{x})^3}{n}, \mu_2 = \frac{\sum (x - \bar{x})^2}{n}$$

$$\text{Coefficient of kurtosis using percentiles (K)} = \frac{Q_3 - Q_1}{2(P_{90} - P_{10})}$$

$$\text{Coefficient of kurtosis using moments } (\beta_2) = \frac{\mu_4}{\mu_2^2}$$

$$\mu_4 = \frac{\sum (x - \bar{x})^4}{n}, \mu_2 = \frac{\sum (x - \bar{x})^2}{n}$$

Practical 2.

Marks secured students of section A and section B in a unit test of Statistics are found as follows;

Section A: 12 , 10 , 9 , 13 , 5 , 14 , 17 , 11 , 3 , 15 , 17 , 19 , 10 , 14 , 6 , 12, 13 , 17

Section B: 7 , 11, 18, 10 , 15 , 17 , 11 , 13 , 15 , 12 , 9 , 14 , 13 , 11 , 14 , 15 , 18 , 12

Find

- Which section is better
- Which section has intelligent students
- Which section has consistent marks
- Which section has more skewed marks

Working formula:

Let x and y be any two variable taking values

$$\bar{x} = \frac{\sum x}{n} \text{ and } \bar{y} = \frac{\sum y}{n}$$

Having more mean is better

Md of x = $(n + 1)/2^{th}$ item of x data after arranging data in ascending order

Md of y = $(n + 1)/2^{th}$ item of y data after arranging data in ascending order

Having more median is intelligent

$$\sigma_x = \sqrt{\frac{1}{n} \sum (x - \bar{x})^2} = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$$\sigma_y = \sqrt{\frac{1}{n} \sum (y - \bar{y})^2} = \sqrt{\frac{\sum y^2}{n} - \left(\frac{\sum y}{n}\right)^2}$$

$$CV_x = \frac{\sigma_x}{\bar{x}} * 100\%$$

$$CV_y = \frac{\sigma_y}{\bar{y}} * 100\%$$

Having less CV is more consistent

Coefficient of skewness using moments $(Y_{1(x)}) = \frac{\mu_{3(x)}}{\mu_{2(x)}^{3/2}}$

$$\mu_{3(x)} = \frac{\sum (x - \bar{x})^3}{n}, \mu_{2(x)} = \frac{\sum (x - \bar{x})^2}{n}$$

Coefficient of skewness using moments $(Y_{1(y)}) = \frac{\mu_{3(y)}}{\mu_{2(y)}^{3/2}}$

$$\mu_{3(y)} = \frac{\sum (y - \bar{y})^3}{n}, \mu_{2(y)} = \frac{\sum (y - \bar{y})^2}{n}$$

Practical 3.

A buyer obtained electric fan from two companies A and B and tested length of life in lab and results are found as

Length of life(hrs)	600 – 800	800 – 1000	1000 – 1200	1200 – 1400	1400 – 1600
No. of electric fans of company A	20	32	52	20	16
No. of fans of company B	6	20	84	24	6

Determine (i) fan of which company has better life (ii) fan of which company has uniform life?(ii) fan of which company has more skewed life

Working formula:

Let x be variable with frequencies in two groups A and B denoted by f_A and f_B respectively

$$\bar{x}_A = \frac{\sum f_A x}{N_A} \text{ and } \bar{x}_B = \frac{\sum f_B x}{N_B}, \text{ where x is mid value}$$

$$\sigma_A = \sqrt{\frac{1}{N_A} \sum f_A (x - \bar{x}_A)^2} = \sqrt{\frac{\sum f_A x^2}{N_A} - \left(\frac{\sum f_A x}{N_A}\right)^2}$$

$$\sigma_B = \sqrt{\frac{1}{N_B} \sum f_B (x - \bar{x})^2} = \sqrt{\frac{\sum f_B x^2}{N_B} - \left(\frac{\sum f_B x}{N_B}\right)^2}$$

$$CV_A = \frac{\sigma_A}{\bar{x}_A} * 100\%$$

$$CV_B = \frac{\sigma_B}{\bar{x}_B} * 100\%$$

Having less CV is more uniform

$$Y_{1(A)} = \frac{\mu_{3(A)}}{\mu_{2(A)}^{3/2}}$$

$$\mu_{3(A)} = \frac{\sum f_A (x - \bar{x}_A)^3}{N_A}, \mu_{2(A)} = \frac{\sum f_A (x - \bar{x}_A)^2}{N_A}$$

$$Y_{1(B)} = \frac{\mu_{3(B)}}{\mu_{2(B)}^{3/2}}$$

$$\mu_{3(B)} = \frac{\sum f_B (x - \bar{x}_B)^3}{N_B}, \mu_{2(B)} = \frac{\sum f_B (x - \bar{x}_B)^2}{N_B}$$

Practical 4.

Following are marks secured by students of a college in Mathematics & Statistics in final exam

Student	Marks in Mathematics	Marks in Statistics
1	45	52
2	30	25
3	51	54
4	32	46
5	17	26
6	45	32
7	28	11
8	20	35
9	46	49
10	11	15
11	56	39

12	47	55
13	28	32
14	33	43

- a. Determine Mean, median, mode, minimum, maximum, quartiles of marks of students in Mathematics
- b. Determine Range, Coefficient of range, interquartile range, semi interquartile range(quartile deviation), coefficient of quartile deviation, standard deviation, coefficient of standard deviation, variance, coefficient of variation, skewness, kurtosis of marks of students in Mathematics
- c. Determine Mean, median, mode, minimum, maximum, quartiles of marks of students in Statistics
- d. Determine Range, Coefficient of range, interquartile range, semi interquartile range(quartile deviation), coefficient of quartile deviation, standard deviation, coefficient of standard deviation, variance, coefficient of variation, skewness, kurtosis of marks of students in Statistics
- e. Determine in which subject students have (i) better marks (ii) consistent marks (iii) less skewed marks (iv) more peaked marks.

Working formula:

Let x be a variable taking different values

$$\text{Mean}(\bar{x}) = \frac{\sum x}{n}$$

Median = $(n + 1)/2^{th}$ item of data after arranging data in ascending order or descending order

Mode = most repeated value of x

Maximum = maximum value of x

Minimum = minimum value of x

Quartiles (Q_i) = $i(n + 1)/4^{th}$ item of data after arranging data in ascending order, $i = 1, 2, 3$

$$\text{Range} = X_{\max} - X_{\min}$$

$$\text{Coefficient of range} = (X_{\max} - X_{\min}) / (X_{\max} + X_{\min})$$

$$\text{Interquartile range} = (Q_3 - Q_1)$$

$$\text{Semi interquartile range} = (Q_3 - Q_1)/2$$

$$\text{Coefficient of quartile deviation} = (Q_3 - Q_1)/(Q_3 + Q_1)$$

$$\text{Standard deviation } (\sigma) = \sqrt{\frac{1}{n} \sum (x - \bar{x})^2} = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$$\text{Coefficient of SD} = \frac{\sigma}{\bar{x}}$$

$$\text{Variance } (\sigma^2) = SD^2$$

$$\text{Coefficient of variation (CV)} = \frac{\sigma}{\bar{x}} * 100\%$$

$$\text{Coefficient of skewness using moments } (\gamma_1) = \frac{\mu_3}{\mu_2^{3/2}}$$

$$\mu_3 = \frac{\sum (x - \bar{x})^3}{n}, \mu_2 = \frac{\sum (x - \bar{x})^2}{n}$$

$$\text{Coefficient of kurtosis using moments } (\beta_2) = \frac{\mu_4}{\mu_2^2}$$

$$\mu_4 = \frac{\sum (x - \bar{x})^4}{n}, \mu_2 = \frac{\sum (x - \bar{x})^2}{n}$$

Practical 5.

Following data represents marks distribution of students;

Marks	Below 10	10 – 30	30 – 60	60 – 80	80 and above
No of students	5	25	30	18	2

Calculate (i) average marks (ii) dispersion of marks (iii) skewness of marks (iv) kurtosis of marks (v) Minimum marks of top 15% students (vi) maximum marks of bottom 20% students (vii) limit of marks of middle 50 % students

Working formula:

$$P_i = L + \frac{\frac{iN}{100} - cf}{f} * h ,$$

$$P_{50} = Md, P_{25} = Q1, P_{75} = Q3$$

$$QD = \frac{Q3-Q1}{Q3+Q1}, S_{KB} = \frac{Q3+Q1-2Md}{Q3-Q1}, K = \frac{Q3-Q1}{2(P90-P10)}$$

Practical 6.

Following data represents marks secured by BSc CSIT students of a college in mid-term exam in Statistics and Mathematics

Student	A	B	C	D	E	F	G	H	I
Marks in Statistics	24	15	19	20	9	17	26	18	8
Marks in Mathematics	18	20	15	22	16	13	24	17	6

Find

- Scatter plot
- Relationship between marks in Statistics and marks in Mathematics using Karl Pearson's method
- Test the significance of the relationship
- Coefficient of determination
- Limit of population correlation coefficient
- Relationship between marks in Statistics and marks in Mathematics using spearman's method

Working formula:

Let x and y be any two variables then relationship between them is given by

$$r = \frac{Cov(x,y)}{\sigma_x \sigma_y} = \frac{\frac{1}{n} \sum (x - \bar{x})(y - \bar{y})}{\sqrt{\frac{1}{n} \sum (x - \bar{x})^2} \sqrt{\frac{1}{n} \sum (y - \bar{y})^2}} = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$\text{Probable error } \{PE(r)\} = 0.6745 \frac{(1-r^2)}{\sqrt{n}}$$

If $r < PE(r)$ then relationship is insignificant

If $r > 6PE(r)$ then relationship is significant

Otherwise no conclusion on significance of relationship

Coefficient of determination (R^2) = r^2

Limit of population correlation coefficient = $r \pm PE(r)$

Spearman's rank correlation coefficient (R) = $1 - \frac{6 \sum d^2}{n(n^2-1)}$

d = difference of rank

Practical 7. Information related to age and monthly salary of workers in a software company are as follows;

Age(yrs)	36	48	29	54	25	57	40	24	33
Monthly salary(000 Rs)	45	54	32	65	29	47	33	35	40

Find

- Suitable regression equation**
- Monthly salary of a worker aged 45 years**
- residuals**
- Coefficient of determination and interpret**
- Standard error of estimate**

Working formula:

Let y be dependent variable and x be independent variable then regression equation of y on x is $y = a + bx$

In equation to find a and b

Solve equations

$$\sum y = na + b \sum x \quad \dots(i) \text{ and}$$

$\sum xy = a \sum x + b \sum x^2 \quad \dots(ii)$ then get a and b and substitute values in the equation to get $\hat{y} = a + bx$

Residuals is the difference between actual value of y and estimated value of $y(\hat{y})$ and is given by $e = y - \hat{y}$

$$\text{Total sum of square (TSS)} = \sum (y - \bar{y})^2 = \sum y^2 - n\bar{y}^2$$

$$\text{Sum of square due to error (residual) (SSE)} = \sum y^2 - a \sum y - b \sum xy$$

Sum of square due to regression (SSR) = TSS – SSE

Coefficient of determination (R^2) = SSR/TSS

Standard error of estimate (S_{yx}) = $\sqrt{SSE/(n - k - 1)}$, SSE = sum of square due to error,

k = number of independent variables in regression equation

Practical 8.

Following are height(ft) and weight(kg) of students of a college in a class

Height(ft)	5.3	4.9	6.0	6.1	5.7	4.8	5.9	4.7	5.4	5.2	4.9	5.4	5.6	5.8
Weight(kg)	55	60	71	69	65	59	74	56	63	58	62	67	73	75

- Draw scatter plot
- Determine relationship between height and weight of students
- Test the significance of the relationship
- Determine coefficient of determination
- Find out limit of population correlation coefficient
- Obtain suitable linear regression equation
- Estimate weight of student having height 6 ft.
- Obtain residuals

Working formula:

Let y be dependent variable and x be independent variable then regression equation of y on x is $y = a + bx$

In equation to find a and b

Solve equations

$\sum y = na + b\sum x$ (i) and

$\sum xy = a\sum x + b\sum x^2$ (ii) then get a and b and substitute values in the equation to get $\hat{y} = a + bx$

Residuals is the difference between actual value of y and estimated value of $y(\hat{y})$ and is given by $e = y - \hat{y}$

Total sum of square (TSS) = $\sum (y - \bar{y})^2 = \sum y^2 - n\bar{y}^2$

Sum of square due to error (residual) (SSE) = $\sum y^2 - a \sum y - b \sum xy$

Sum of square due to regression (SSR) = TSS – SSE

Coefficient of determination (R^2) = SSR/TSS

Standard error of estimate (S_{yx}) = $\sqrt{SSE/(n - k - 1)}$, SSE = sum of square due to error,

k = number of independent variables

in regression equation

Practical 9.

From following information related to weight of employees working in a software company an employee is selected at random then find probability that employee has weight (i) Between 60 to 80 kg (ii) less than 70 kg (iii) At least 80 kg

Weight (kg)	Number of employees
40 – 50	3
50 – 60	12
60 – 70	23
70 – 80	17
80 – 90	8
90 – 100	2

Working formula

Let A be any event then probability of occurrence of event A is

$P(A) = m/N$, m = favorable number of cases for event A and

N = exhaustive number of cases for event A

Practical 10. Following information is obtained from a survey related to brand of laptop use according to age group of people. A person is selected at random then (i) what is probability that person has Lenovo laptop?(ii) what is probability that person is of age group 35-45 given that uses Dell laptop? (iii) What is probability that person is of age group 25 – 35 and uses Acer laptop? (iv) What is probability that person uses Samsung laptop given that of age group 55- 65?(v) What is probability that person is of age group 55 – 65 given that uses hp laptop?

				Laptop brand		
		Acer	Lenovo	Dell	hp	Samsung
	15-25	12	21	18	11	14
Age group	25-35	20	18	22	9	12
	35-45	23	10	18	13	9
	45-55	8	16	13	7	14
	55-65	7	8	9	5	6

Working formula

Let A be any event then probability of occurrence of event A is

$P(A) = m/N$, m = favorable number of cases for event A and

N = exhaustive number of cases for event A

For any two dependent events A and B

$$P(A/B) = P(A \cap B) / P(B) = m_{AB} / m_B ,$$

m_{AB} = favorable number of cases for both events A and B

m_B = favorable number of cases for event B

Practical 11.

Following information represents student number and result of bachelor level student under different programs. Find (i) what is probability that a student selected at random pass exam? (ii) what is probability that a student selected at random pass exam is Engineering student (iii) what is probability that a student selected at random fail exam is education student.

Program	Student number	Pass percentage
Science	4500	45
Management	85000	30
Humanities	9000	40
Law	7500	20
Education	28000	15

Engineering	7000	60
Medicine	10000	75
Agriculture	1500	80
Forestry	1000	85

Working formula

Let $A_1, A_2, A_3 \dots \dots \dots A_n$ be mutually exclusive and exhaustive events defined on sample space and B be any other event defined which subset of sample space then

$$P(A_i/B) = P(A_i)P(B/A_i) / P(B), i = 1, 2, 3, \dots, n$$

$$P(B) = \sum_{i=1}^n P(A_i)P(B/A_i)$$

Practical 12.

On rolling two dice find (i) mean (ii) SD (iii) Skewness and (iv) Kurtosis of sum of faces on dice

Working formula:

$$\text{Total outcomes} = 6^n$$

Let X = sum of faces on dice and $p(x)$ be probability mass function then

$$\mu'_1 = \sum XP(X)$$

$$\mu'_2 = \sum X^2P(X)$$

$$\mu'_3 = \sum X^3P(X)$$

$$\mu'_4 = \sum X^4P(X)$$

$$\mu_2 = \mu'_2 - \mu'^2_1$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3$$

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

$$\text{Mean} = \mu'_1$$

$$\text{SD} = \sqrt{\mu_2'}$$

$$\text{Skewness} = \gamma_1 = \mu_3 / \mu_2'^{3/2}$$

$$\text{Kurtosis} = \beta_2 = \mu_4 / \mu_2'^2$$

Practical 13.

Following data represents marks obtained by student in a chapter test. Form probability distribution and find (i) moments about origin (ii) moments about mean (iii) measure of central tendency (iv) measure of dispersion (v) measure of skewness (vi) measure of kurtosis of marks.

Marks secured	Number of students
1	1
2	2
3	4
4	7
5	9
6	8

7	6
8	5
9	4
10	2

Working formula

Let X = marks secured and $p(x)$ be probability mass function then

Moments about origin

$$\mu'_1 = \sum XP(X)$$

$$\mu'_2 = \sum X^2P(X)$$

$$\mu'_3 = \sum X^3P(X)$$

$$\mu'_4 = \sum X^4P(X)$$

Moments about mean

$$\mu_1 = 0$$

$$\mu_2 = \mu'_2 - \mu_1'^2$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3$$

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2\mu_1'^2 - 3\mu_1'^4$$

Central tendency = Mean = μ'_1

Dispersion = SD = $\sqrt{\mu_2}$

Skewness = $\gamma_1 = \mu_3/\mu_2^{3/2}$

Kurtosis = $\beta_2 = \mu_4/\mu_2^2$

Practical 14.

From following information of students of a college according to height and weight obtain (i) Joint probability (ii) Marginal probability (iii) Marginal probability distribution function

Weight(kg) Height(ft)	40 – 50	50 – 60	60 – 70	70 – 80
4 – 4.5	2	5	1	0
4.5 – 5	7	29	12	2
5 – 5.5	3	42	31	9
5.5 – 6	1	13	52	31

Working formula:

For any two random variable x and y joint probability is $P(x,y) = \frac{f(x,y)}{N}$

Where $N = \sum f(x,y)$

For any two random variable x and y marginal probability of random variable X is $P(x) = \sum_y P(x,y)$

For any two random variable x and y marginal probability of random variable Y is $P(Y) = \sum_x P(x,y)$

For any two random variable x and y marginal probability distribution function of random variable X is $F(x) = P(X \leq x)$

For any two random variable x and y marginal probability distribution function of random variable Y is $F(y) = P(Y \leq y)$

Practical 15. Fit binomial distribution to the data given below;

x:	0	1	2	3	4	5	6	7	8
f:	17	20	32	41	56	51	39	22	12

Find (i) $P(x)$ (ii) $F(x)$ (iii) $P(X \geq 2)$ (iv) $P(X < 5)$ (v) $P(X = 7)$ (vi) expected frequencies

Working formula:

Let $x \sim B(n, p)$ then $p(x) = c(n, x) p^x q^{n-x}$

Where n and p are parameters

To find p

$$E(x) = np$$

Also $E(x) = \bar{x} = \sum fx / N$, where $N = \sum f$

$$\text{Hence } np = \bar{x}$$

$$\text{Or, } p = \bar{x}/n$$

$$\text{Expected frequency} = Np(x)$$

Practical 16. Fit Poisson distribution to following data

x:	0	1	2	3	4	5	6	7
f:	60	81	92	104	78	61	55	40

Find (i) $P(x)$ (ii) $F(x)$ (iii) $P(X > 4)$ (iv) $P(X \leq 5)$ (v) $P(X=4)$ (vi) Expected frequency

Working formula:

$$\text{Let } x \sim P(\lambda) \text{ then } p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Where λ parameter

To find λ

$$E(x) = \lambda$$

Also $E(x) = \bar{x} = \sum fx / N$, where $N = \sum f$

Hence $\lambda = \bar{x}$

Expected frequency = $Np(x)$

Practical 17.

Fit normal distribution to following data

Class:	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90	90 – 100
Frequency:	3	15	26	34	28	19	11	4

Find (i) $P(30 < X < 70)$ (ii) $P(X > 60)$ (iii) $P(X < 80)$ (iv) expected frequency

Working formula:

Let $x \sim N(\mu, \sigma^2)$ then $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2(\frac{x-\mu}{\sigma})^2}$

Where μ and σ^2 are parameters

To find μ and σ^2

$\mu = \bar{x} = \sum fx / N$, where $N = \sum f$

$$\sigma^2 = \frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2$$

$Z = \frac{x-\mu}{\sigma}$ then $f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$

$$F(z_i) = \int_{-\infty}^{z_i} f(z) dz$$

$$P(z_i < Z < z_{i+1}) = F(z_{i+1}) - F(z_i)$$

Expected frequency = $N P(z_i < Z < z_{i+1})$

Practical 18.

Fit negative exponential distribution to following data

Decay time: 0 – 0.5 0.5 – 1.0 1.0 – 1.5 1.5 – 2.0 2.0 – 2.5

Number of particles: 44 36 30 19 13

Find (i) $P(0.5 < X < 2.0)$ (ii) $P(X > 1.5)$ (iii) $P(X < 1.0)$ (iv) expected frequency

Working formula:

Let $x \sim \exp(\theta)$ then $f(x) = \theta e^{-\theta x}$

Where θ is parameter of the distribution

To find θ ,

$$E(x) = \frac{1}{\theta}$$

$$E(x) = \bar{x} = \frac{1}{\theta}$$

$$\text{Or, } \theta = \frac{1}{\bar{x}}$$

$$\bar{x} = \sum fx / N, \text{ where } N = \sum f$$

$$P(a < x < b) = \int_a^b f(x) dx$$

$$\text{Expected frequency} = N P(a < x < b)$$